

## packages/graph-view/src/utils/elk-layout.worker.ts

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### Overview

A new worker script `packages/graph-view/src/utils/elk-layout.worker.ts` (lines 1-51) adds off-main-thread ELK layout. The file header (R1-3) states it keeps the UI responsive on large knowledge graphs.

### Key changes

- **Message types:**
  - `ElkWorkerRequest` (R6-9)
  - `ElkWorkerSuccess` (R11-15)
  - `ElkWorkerFailure` (R17-21)
  - `ElkWorkerResponse` union (R23)
- **Lazy ELK loading:** `getElk()` (R27-31) caches a `Promise<ElkLayoutInstance>` created via `loadElkConstructor()` imported from `./elk-bundled` (R4).
- **Worker loop:** `self.onmessage` (R34-51) receives `ElkWorkerRequest`, runs `elk.layout(input)`, and posts back either `ElkWorkerSuccess` or `ElkWorkerFailure`. Errors are stringified and returned in `ElkWorkerFailure`.

### Impact

- **Execution context:** Layout runs inside a Web Worker, isolating heavy computation from the main thread.
- **Type safety:** Typed interfaces reduce runtime message-handling bugs.
- **Build:** The worker is a separate bundle; it must be emitted by the build system and referenced by the application.

### Risks & follow-ups

- **Worker registration:** Ensure the worker script is correctly referenced in the application entry point and emitted to the expected location.
- **ELK constructor loading:** `loadElkConstructor()` must resolve to a valid constructor; otherwise layout requests will fail.
- **Concurrency:** Multiple requests share the same `elkPromise`; confirm that this does not cause race conditions or stale state.
- **Error surface:** `err instanceof Error` is used to extract the message; unknown from the available diff/scan evidence whether all error types are handled correctly.